Regression output – Railways and happy danes

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1 Questions

2 Method

2.1 Models

Estimating the DiD-equation:

$$\Delta Y_p = \theta_0 + \alpha_c + R_p + \mathbf{X}_p + \epsilon_p$$

- ΔY_p : Changes in outcome variable from 1850-1901 (population etc.)
- θ_0 : Constant term (not 100% if it should be there)
- α_c : County fixed effects
- R_p : Rail variable based on distance to rail or connectivity in 1875
- $\mathbf{X}_p$: Controls including: log_area_parish, boulder_clay_pct log_dist_coast, log_distance_market_town, log_distoxroad, log_dist_hmb, log_dist_cph

Constant or no constant? Looks like no constant in Berger 2019 eq 1, however does this make sence? In my understanding including a constant in the simple regression: $\Delta Y_p = \theta_0 + \alpha_c$, the interpretation of the constant would be the average change in the outcome variable of the ommited county (ie. $E(\Delta Y_p | \alpha_c = 0)$).

2.2 Restrictions:

Restricting on less than 1 km to market town.

3 Descriptives

	No trunk line access		Trunk line access		Total	
	Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.
pop_1850	667.21	462.39	732.72	563.67	671.30	469.41
pop_1901	878.49	621.12	1,119.70	901.20	893.56	644.46
d_log_pop	0.27	0.29	0.38	0.42	0.28	0.30
log_area_parish	9.80	0.68	9.72	0.62	9.80	0.68
Boulder_clay_pct	0.48	0.32	0.50	0.32	0.48	0.32
log_dist_coast	8.61	1.02	8.66	0.87	8.62	1.01
log_distance_market_town	2.29	0.55	1.92	0.71	2.27	0.57
log_distoxroad	3.67	1.38	3.81	1.30	3.68	1.38
log_dist_hmb	5.62	0.25	5.56	0.22	5.62	0.25
log_dist_cph	4.94	0.66	4.81	0.68	4.93	0.66
Observations	1471		98		1569	

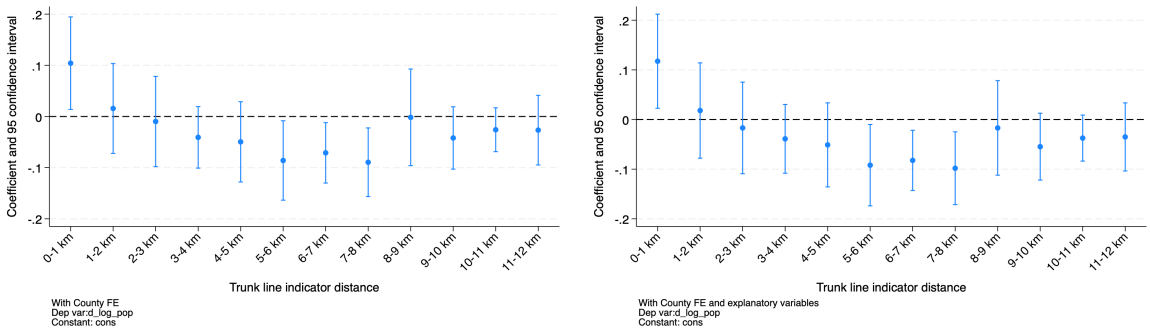
Table 1: Descriptives 1km ind

	No trunk line access		Trunk line access		Total	
	Mean	Std.dev.	Mean	Std.dev.	Mean	Std.dev.
pop_1850	672.03	468.21	666.66	478.04	671.30	469.41
pop_1901	881.76	628.51	968.29	734.71	893.56	644.46
d_log_pop	0.27	0.28	0.35	0.37	0.28	0.30
log_area_parish	9.81	0.69	9.70	0.61	9.80	0.68
Boulder_clay_pct	0.47	0.32	0.49	0.33	0.48	0.32
log_dist_coast	8.59	1.03	8.78	0.82	8.62	1.01
log_distance_market_town	2.31	0.55	1.98	0.64	2.27	0.57
log_distoxroad	3.66	1.40	3.80	1.20	3.68	1.38
log_dist_hmb	5.63	0.25	5.57	0.24	5.62	0.25
log_dist_cph	4.94	0.66	4.87	0.67	4.93	0.66
Observations	1355		214		1569	

Table 2: Descriptives 2km ind

4 Population

4.1 Bin regressions



(a) (b)

Figure 1: Rail distance indicators (1km bins) on ln population

4.2 OLS

	Δ (log)Population 1850-1901		
	(1)	(2)	(3)
Rail pre-1876 (connected)	0.083***	0.090***	0.115***
	(0.020)	(0.019)	(0.018)
Rail 1876-1901 (connected)			0.092***
			(0.017)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
R^2	0.287	0.319	0.334
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Δ (log)Population 1850-1901		
	(1)	(2)	(3)
Rail pre-1876 (< 1km)	0.132**	0.150***	0.159***
	(0.035)	(0.035)	(0.035)
Rail 1876-1901 (< 1km)			0.102***
			(0.021)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
R^2	0.287	0.321	0.330
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Δ (log)Population 1850-1901		
	(1)	(2)	(3)
Rail pre-1876 (< 2km)	0.088**	0.104***	0.124***
	(0.022)	(0.023)	(0.022)
Rail 1876-1901 (< 2km)			0.103***
			(0.019)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
R^2	0.286	0.320	0.336
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Δ (log)Population 1850-1901		
	(1)	(2)	(3)
log_dist_rail_1875	-0.017	-0.023	-0.031*
	(0.010)	(0.012)	(0.011)
Rail 1876-1901 (< 2km)			0.102***
			(0.018)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
R^2	0.280	0.312	0.328
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: OLS

4.3 IV

4.3.1 First stage

	Rail pre-1876 (< 1 km)		
	(1)	(2)	(3)
dist_rail_1875_inst_i_1	0.162**	0.139**	0.138**
	(0.048)	(0.045)	(0.045)
Rail 1876-1901 (< 1km)			-0.059***
			(0.010)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
Kleibergeren-Paap F-statistic	11.68	9.44	9.53
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	Rail pre-1876 (< 2 km)		
	(1)	(2)	(3)
dist_rail_1875_inst_i_2	0.286***	0.244***	0.229***
	(0.043)	(0.046)	(0.045)
Rail 1876-1901 (< 2km)			-0.127***
			(0.017)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
Kleibergeren-Paap F-statistic	44.41	28.46	26.40
Observations	1569	1569	1569

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: First stage

4.3.2 IV results

	Δ (log)Population 1850-1901		
	(1) d_log_pop	(2) d_log_pop	(3) d_log_pop
Rail pre-1876 (< 1km)	0.104	0.163	0.182
	(0.183)	(0.206)	(0.202)
Rail 1876-1901 (< 1km)			0.103***
			(0.026)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
Kleibergeren-Paap F-statistic			
Observations	1569	1569	1569

Standard errors in parentheses
Instrument: dist_rail_1875_inst_i_1
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

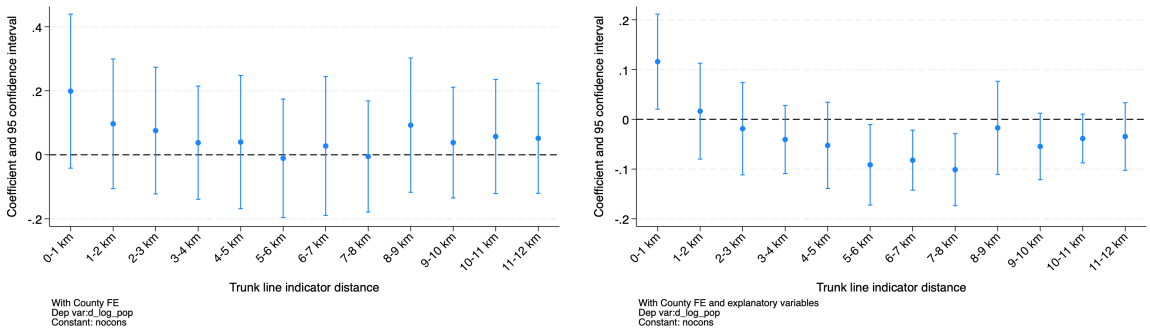
	Δ (log)Population 1850-1901		
	(1) d_log_pop	(2) d_log_pop	(3) d_log_pop
Rail pre-1876 (< 2km)	0.053	0.077	0.126
	(0.071)	(0.074)	(0.076)
Rail 1876-1901 (< 2km)			0.104***
			(0.023)
County FE	Yes	Yes	Yes
Additional controls	No	Yes	Yes
Kleibergeren-Paap F-statistic			
Observations	1569	1569	1569

Standard errors in parentheses
Instrument: dist_rail_1875_inst_i_2
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: IV results

5 Appendix

5.1 Population without constant



(a) (b)

Figure 2: Rail distance indicators (1km bins) on ln population (nocons)

Table 6: OLS 2km (with constant)

5.2 Rail indicators

	0 b/rowpct	1 b/rowpct	Total b/rowpct
Aalborg	97	12	109
	88.99083	11.00917	100
Aarhus	49	12	61
	80.32787	19.67213	100
Bornholm	15	0	15
	100	0	100
Frederiksborg	52	4	56
	92.85714	7.142857	100
Hjoerring	70	8	78
	89.74359	10.25641	100
Holbaek	69	13	82
	84.14634	15.85366	100
Koebenhavn	55	18	73
	75.34247	24.65753	100
Maribo	69	21	90
	76.66667	23.33333	100
Odense	88	15	103
	85.43689	14.56311	100
Praestoe	68	12	80
	85	15	100
Randers	121	9	130
	93.07692	6.923077	100
Ribe	53	18	71
	74.64789	25.35211	100
Ringkoebing	88	16	104
	84.61538	15.38462	100
Skanderborg	54	11	65
	83.07692	16.92308	100
Soroe	69	12	81
	85.18519	14.81481	100
Svendborg	82	2	84
	97.61905	2.380952	100
Thisted	64	0	64
	100	0	100
Vejle	57	13	70
	81.42857	18.57143	100
Viborg	135	18	153
	88.23529	11.76471	100
Total	1355	214	1569
	86.36074	13.63926	100
Observations	1569		

(a) Data from project

	0 b/rowpct	1 b/rowpct	Total b/rowpct
Frederiksborg	51	4	55
	92.72727	7.272727	100
Hjørring	73	8	81
	90.12346	9.876543	100
Holbæk	69	13	82
	84.14634	15.85366	100
København	50	24	74
	67.56757	32.43243	100
Maribo	91	0	91
	100	0	100
Odense	89	15	104
	85.57692	14.42308	100
Præstø	70	10	80
	87.5	12.5	100
Randers	123	9	132
	93.18182	6.818182	100
Ribe	54	15	69
	78.26087	21.73913	100
Ringkøbing	94	17	111
	84.68468	15.31532	100
Skanderborg	55	10	65
	84.61538	15.38462	100
Sorø	70	14	84
	83.33333	16.66667	100
Svendborg	84	2	86
	97.67442	2.325581	100
Thisted	95	0	95
	100	0	100
Vejle	61	14	75
	81.33333	18.66667	100
Viborg	133	18	151
	88.07947	11.92053	100
Ålborg	100	11	111
	90.09009	9.90991	100
Århus	49	12	61
	80.32787	19.67213	100
Total	1411	196	1607
	87.80336	12.19664	100
Observations	1607		

(b) Data used in thesis

Table 7: Rail <2km in 1875 by county
State only for thesis which differ in Lolland/Falster